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In the above declaration, data-type refers to any valid C++ data-type, array_name refers to the name of the 2-D array, row-size refers to the number of rows and col-size refers to the number of columns in the 2-D array.

For example

```
int A[3][4];
```

In the above example, A is a 2-D array, 3 denotes the number of rows and 4 denotes the number of columns. This array can hold a maximum of 12 elements.

Note

Array size must be an unsigned integer value which is greater than 0. In arrays, column size is compulsory but row size is optional.

Other examples of 2-D array are:

```
int A[3][3];

float x[2][3];

char name[5][20];
```

12.3.2 Initialization of Two-Dimensional array

The array can be initialized in more than one way at the time of 2-D array declaration.

For example

```
int matrix[4][3]={
{10,20,30},// Initializes row 0
{40,50,60},// Initializes row 1
{70,80,90},// Initializes row 2
{100,110,120}// Initializes row 3
};
int matrix[4][3]={10,20,30,40,50,60,70,80,90,100,110,120};
```

Array's row size is optional but column size is compulsory.

For example